ENGN 2350: Data-Driven Design & Analysis of Structures & Materials (3dasm)

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Instructor Office Hours: See Canvas course page

Course webpage: Github Repository
Class Hours: MWF (3×50 min)
Class Room: See Courses @ Brown

Course Description

This course focuses on the data-driven design and analysis of structures and materials. The aim is to balance theory and practice, such that students understand and use data-driven methods in new scenarios. The first half of the course introduces machine learning from a probabilistic perspective, providing a foundation to understand current machine learning methods. The second part of the course focuses on applying the data-driven strategy to solve engineering problems. Although most examples covered will target solid mechanics applications, the content is applicable to different fields. The course is primarily lecture driven, but the lecture slides are interactive and include Python code that runs live.

Prerequisites

- Working knowledge of college-level calculus and linear algebra is required. Courses equivalent to MATH 0190, MATH 0200, and MATH 0520.
- Basic programming knowledge, preferably in Python, is required. However, a course equivalent to CSCI 0150 or CSCI 0170 is sufficient.

Prior coursework in statistics or probability (e.g. APMA 1650) is beneficial, but not strictly needed.

Required Textbooks

- Murphy, Kevin P. *Probabilistic machine learning: an introduction*. MIT press, 2022. Available online.

Recommended Textbooks

- Bishop, Christopher M. *Pattern recognition and machine learning*. Springer Verlag, 2006.
- Géron, Aurélien. *Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow*. O'Reilly Media, Inc., 2022.

Course Objectives

Upon completion of this course, students should be able to:

1. Understand, derive and analyze machine learning methods from a probabilistic perspective;
2. Understand basic sampling methods (design of experiments);
3. Understand basic optimization algorithms;
4. Identify appropriate sampling, data generation, machine learning, and optimization methods based on the application of interest;
5. Develop data-driven strategies for new problems.

Course Policies and Overall Expectations

Students in this course will be expected to do the following:

1. Attend all lectures and actively participate in discussion;
2. **Optionally**, bring laptop to the lectures because slides include interactive Python code that can be run live (however, bringing laptop is not required);
3. Read all assigned materials as indicated at the beginning of each lecture.
4. Complete and turn in all assignments on time. Solutions to homework must be clearly written with appropriate tables and figures included.
5. Demonstrate an understanding of course content on the midterm examination and the final project.
6. Respect each other, including questions posed by others and corresponding discussion.

Evaluation

Students will be evaluated based on:

Grade Category	Percentage
Homework	10%
Midterm Exam	40%
Final Project	50%

Evaluation Category Details

Homework. Homework assignments will be graded only with 5 levels: A+ (100%; fully correct), A (90%; has minor error), B (75%; has significant error or several minor errors), C (60%; several significant errors), D (0%, homework not delivered). If you deliver something with an honest attempt at solving the homework you are assigned 60% for that homework. Late Homework can only get up to A (90%). The worst Homework is removed.

Midterm Exam. Midterm will be in class, and will cover materials presented in lectures, readings and assignments. The midterm is closed-book. As a general rule, make up exams and advance exams will not be given, with the exception when the dates conflict with religious observances or truly exceptional circumstance. Please let me know ahead of time if you need to make alternative arrangements for this purpose.

Final Project. The Final Project involves using the data-driven strategy to solve a simple Engineering problem. The project deliverables include a mandatory report (PDF) and the necessary code to generate all the results (a single ZIP folder). The report needs to explain the methods used, the reasoning behind method selection, and results. The code needs to be properly commented and it must be possible to run it without modifications. Projects are expected to be done individually, but the instructor may allow teams composed of no more than two members.

Use of LLMs and AI coding agents. Students are encouraged to use large language models (ChatGPT, Claude, Gemini, open-weight models, etc.) and AI coding agents (Claude Code, opencode, Copilot, etc.) throughout the course, including for homework. However, be aware that **using an LLM to do your homework can significantly impair your learning ability**: two large controlled studies found that students with AI help scored much better on homework and then **scored worse than students without AI on the exams** that followed [1, 2]. This is why:

1. Homeworks only weigh 10% of the grade because good LLMs can solve them.
2. The midterm is closed-book (no computer, no Internet, and... no LLM)
3. The Final Project has an open-ended component. Working with an LLM is encouraged. The question is: how great can that project become?

Whatever tools students use, they must be able to **explain and defend every part of their code and report during the presentation**; copying from other students or groups remains plagiarism and results in a grade of zero.

Time Allocation

This is a one credit course, amounting to 180 hours of in-class and out-of-class work over the semester as listed below.

[1]Estimate includes time in office hours which should reduce total time to complete problems, and assumes part of your usual preparation for solving problems is captured in the reading time estimate

Task	Hours Spent on Task
Thirty-eight 50-minute class meetings (lectures, Q&A, midterm, presentations)	32
Reading (lectures and book)	28
Regular homework assignments[1]	36
One midterm exam (in class; 36 h to study)	36
Final Project	48
Total	180

Class Schedule

A detailed schedule is available on the course webpage.

Accessibility and Accommodations Statement

Brown University is committed to full inclusion of all students. Please inform the professor early in the term if you may require accommodations or modification of any course procedures. You may speak with the professor after class, during office hours, or by appointment. If you need accommodations around online learning or in classroom accommodations, please be sure to reach out to Student Accessibility Services (SAS) for their assistance (sas@brown.edu, 401-863-9588). Undergraduates in need of short-term academic advice or support can contact an academic dean in the College by emailing college@brown.edu. Graduate students may contact one of the deans in the Graduate School by emailing graduate_school@brown.edu.

Diversity Statement

This course is designed to support an inclusive learning environment where diverse perspectives are recognized, respected and seen as a source of strength. Our intent is to provide materials and activities that are respectful of various levels of diversity: mathematical background, previous computing skills, gender, sexuality, disability, age, socioeconomic status, ethnicity, race, and culture.

English Language Learners

Brown University welcomes students from around the world, and the unique perspectives international students bring enrich the campus community. To empower students whose first language is not English, an array of ELL support is available on campus including language and culture workshops and individual appointments. For more information about English Language Learning at Brown, contact the ELL Specialists at ellwriting@brown.edu.

References

[1] Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakcı, and Rei Mariman. Generative AI without guardrails can harm learning: Evidence from high school mathematics. *Proceedings of the National Academy of Sciences*, 122(26):e2422633122, 2025. doi: 10.1073/pnas.2422633122.

[2] David Strömberg, Victor Lei, and Yanhui Wu. The generative AI learning penalty: Evidence from Chinese secondary education. CEPR Discussion Paper 21577, Centre for Economic Policy Research, 2026.